Muhammed Raseel P

Flutter Developer | Mobile App Engineer

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ABOUT ME

Flutter Developer with **1.5 years of professional experience** and nearly **3 years of hands-on experience working with Flutter and native Android development in Kotlin**, with strong experience using **GetX** and integrating **REST APIs**, along with testing and debugging using **Postman**. I have **strong native Android experience** building performance-critical, low-level modules including on-device machine learning pipelines and **custom Flutter–Kotlin bridge architectures**, that go well beyond what the Flutter framework handles on its own. I have collaborated closely with **.NET-based backend** systems at work and explore backend development using **Go and MongoDB** through personal projects. Experienced in using **Firebase services** in production to **improve app stability, performance, and reliability**. I have experience releasing and maintaining production-ready cross-platform applications on both the **Play Store and App Store.**

SKILL

- **Programming Languages:** Dart, Kotlin, Python, Go, C
- **Frameworks & Technologies:** Flutter, Android (Native), GetX, Provider, WorkManager, Foreground Services, MediaStore, Gin, Django, REST APIs
- **Databases:** SQLite, MongoDB, MySQL
- **Firebase:** Crashlytics, Performance Monitoring, App Check, Cloud Messaging (FCM)
- **Tools & Platforms:** Play Store & App Store (Production Releases), Postman, Git, Azure DevOps, GitHub, ffmpeg
- **AI / ML Integration:** ONNX Runtime (XNNPACK), On-Device Inference, Model Quantization (int8), PyTorch to ONNX Export, Face Detection & Recognition, MobileCLIP / CLIP Semantic Search

PROFESSIONAL EXPERIENCE

Flutter Developer Full-Time, -- Logiology Solutions 04/2025 – Present | Calicut, Kerala

Worked as a Flutter Developer in a **product-based company delivering Jewellery ERP solutions**, contributing to production-grade mobile applications used in real business environments.

N Vend – Jewellery Sales & Billing Application for Salesmen

- **Maintained and enhanced N Vend**, a core jewellery sales application, with **Play Store** and **App Store deployments**.
- **Designed and developed a complete CRM module**, and automated activity and opportunity creation, scheme and receipt workflows, and customer interaction tracking.
- Worked extensively with **complex mathematical and financial calculations**, including multiple taxation models such as **CGST, SGST, IGST, VAT**, ensuring accuracy and compliance.
- Implemented **PDF generation and printing workflows** using the pdf package, with **custom layouts for invoices and proforma documents.**
- Integrated **Firebase Cloud Messaging (FCM)** to enable real-time notifications for business-critical events.
- Built **MRZ parsing** and **debit/credit card data parsing** features to support identity and payment-related workflows.
- Integrated **RFID-based stock-taking,** enabling bulk scanning of tagged inventory through **radio-frequency identification and real-time item tracking.**

N Divo – Jewellery Business Analytics Application for Owners

- Worked on **N Divo**, a jewellery owner-focused business monitoring application, deployed on the **Play Store** and **App Store**.
- Implemented and enhanced dashboards and reports covering **stock, sales, salesmen performance, profits, assets, liabilities**, and other key business metrics.
- Developed a **branch-level aggregation module**, enabling owners to view **combined data from multiple jewellery branches** in a single, unified view.
- Implemented **biometric authentication** to secure sensitive business and financial data, strengthening application security and access control.
- Integrated **Firebase services** to improve application reliability, monitoring, and real-time updates.

Flutter Developer Intern, -- Regional Technologies 01/2025 – 02/2025 | Calicut, Kerala

- Led a team of 4 developers to create a **real-time emergency vehicle priority system** using Flutter and Django, reducing ambulance response times by **30%.**
- Collaborated on **geofencing notifications, SOS requests, and a medical chatbot**, enhancing user engagement by **20%.**

PERSONAL PROJECTS

Lost & Found: Offline Photo Search & Face Recognition 12/2025 – Present
(Flutter, Kotlin, ONNX Runtime, MobileCLIP-S0, SCRFD, MobileFaceNet, SQLite, WorkManager)

- Built an **open-source Android app** that lets people find any photo or video by **describing it in plain words** (like "birthday cake with candles") or by **who is in it**, running **completely on the phone** with no internet, account or cloud.
- Made the app's **AI search 6x smaller and more accurate** by switching from **OpenAI's CLIP (ViT-B/32)** to Apple's **MobileCLIP-S0** model (download cut from **606 MB to 104 MB**), and reduced the **app size by 67% (398 MB to 131 MB)** by moving AI processing to **ONNX Runtime**.
- Built **on-device face recognition** that finds the people in your photos and **groups them automatically** (**SCRFD** face detection, **MobileFaceNet** recognition), including small faces in group photos, and tuned to run fast on each phone.
- Designed a **smart grouping algorithm** that keeps the same person together even with **glasses, new hairstyles or years apart**, tested on **4,400+ faces**, with **one-tap fixes** for mistakes (merge, split, undo) that the app remembers.
- Created a **self-correcting tool** that spots when **two different people were mixed up as one** and separates them automatically (**multi-seed 2-means clustering**), verified at **100% accuracy on 40 real cases**.
- Extended search and face recognition to **videos**, so users can **jump to the exact moment** a person or scene appears, and combined both features so searches like **"Maya at the beach"** work, while recognising only **1–2 faces per person per clip** to stay fast.
- Built **memory features** people come back to daily: **On this day**, a **timeline of everyone's photos** with key moments (first photo together, reunions), **before-and-after** comparisons and **daily memory notifications**, plus **zip export** of large photo collections.
- Engineered a **reliable background system** that scans thousands of photos without freezing the phone: **Android foreground services and WorkManager**, a **resumable, checksum-verified** model download, and coordination of the phone's limited **video decoders**.
- Designed the **communication layer between the Flutter interface and native Android (Kotlin)**, so the heavy AI work runs natively at full speed while the app stays smooth: **MethodChannels** for search and actions, **EventChannels** streaming **live scan progress**, and support for **searching while a scan is still running**.

iPhone-to-Android Video Compatibility Converter, (Kotlin, Media3 Transformer, FFmpeg) 09/2026

- Built a **native Android app** that fixes iPhone videos that fail to play on Android, converting them into a **universally supported format** by combining several different video processing engines to **maximize compatibility** across devices.
- Designed a **multi-step fallback system** that automatically retries a conversion using a different strategy whenever the first one fails, since real iPhone HDR/Dolby Vision videos trigger inconsistent, hardware-specific failures depending on the device.
- Integrated **FFmpeg** as a backup video engine for the cases where Android's built-in tools couldn't process a file at all, **along with automatic color correction** so HDR footage still looks accurate after conversion.
- Assembled a custom **FFmpeg processing chain** using color-space and tone-mapping filters **(zscale, tonemap)** for accurate HDR-to-standard color conversion, **hardware-accelerated encoding for speed, and metadata correction** for proper video orientation and instant playback on Android.
- Identified and fixed a hardware limitation where some devices could play 4K video but not create it, by having the app **detect each device's real capabilities** and adjust the output resolution to match automatically.

EDUCATION

Bachelor of Computer Science 2022 – 2025 | Malappuram
College of Applied Science Vazhakkad, Malappuram

CERTIFICATES AND ACHIEVEMENTS

BR Hackathon (Organized by PearAI)
PearAI (Y Combinator W24) Track Winner